Tejus Samir

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EDUCATION

Texas A&M University - College Station *May 2029*
Bachelor of Science in Electrical Engineering
Extracurriculars: Ignitors Rocketry Team, Institute of Electrical and Electronics Engineers (IEEE)

EXPERIENCE

Ignitors Rocketry Team, Electronics Mentee - College Station, TX *Jan 2026 – May 2026*

- Trained in rocket avionics and embedded systems, working with microcontrollers, sensors, and serial communication protocols (UART, I2C, SPI) used in flight electronics
- Used OpenRocket to simulate flights and predict key performance metrics, including apogee and parachute deployment speed, applying concepts in propulsion, drag, and flight dynamics
- Used Altium Designer for PCB design and contributed to the build and successful launch of a team rocket

Data Science Intern, Fortune Makers LLC - Plano, TX *Jun 2024 – Aug 2024*

- Collected, organized, and aligned earnings release date data to construct comprehensive financial datasets for analysis
- Researched and implemented statistical regression techniques, including Principal Component Analysis (PCA) and Lasso Regression, to extract trends and reduce dimensionality in financial data
- Utilized Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn for data manipulation, visualization, and exploratory analysis
- Analyzed model outputs to derive insights into stock behavior and improve interpretability of results

PROJECTS

FreshPal Smart Sensor | Altium Designer, ESP32, I2C, Embedded Systems, IoT *Aug 2023 – May 2025*

- Designed and fabricated a custom 2-layer PCB in Altium Designer integrating an ESP32 microcontroller, an I2C temperature/humidity sensor, and a Winsen GM-802B MEMS ammonia (NH_3) sensor to detect food spoilage
- Developed the NH_3 sensing circuitry, including heater drive and voltage-divider signal conditioning, and programmed C++ firmware in Arduino IDE using non-blocking architecture, I2C communication, and Wi-Fi data transmission
- Authored a 22-page feasibility/technical plan that achieved 3rd place internationally at the DECA International Career Development Conference among 2,500+ competitors

Audio Signal Conditioning Amplifier | TINA-TI, Altium Designer *Aug 2026*

- Designed and simulated an inverting op-amp signal-conditioning amplifier in TINA-TI, configuring feedback network component values to achieve a passband gain of 23.28 dB (~14.6 V/V) with an integrated active low-pass anti-aliasing filter stage cutting off at ~1.06 kHz.
- Laid out the circuit as a PCB in Altium Designer, translating the simulated design into a physical board.

Counterfeit Semiconductor Detection | Semiconductor Validation and Testing, Statistical Modeling *Apr 2026*

- Developed a software-based solution to fingerprint and authenticate semiconductors using DAC Differential Nonlinearity (DNL) and Integral Nonlinearity (INL) test data as chip-level identifier
- Applied Control Charts to these linearity signatures, validating against known authentic and counterfeit samples to reliably distinguish genuine chips from counterfeits with 98.7% accuracy
- Presented to a panel of industry judges to win 1st place and a $5,000 grand prize at Aggies Invent: Hardware Resilience

SKILLS AND CERTIFICATIONS

Awards/Honors: Eagle Scout (Scouting America), Aggies Invent - 1st Place, DECA Internationals - 3rd Place
Programming: Python, C++, Java, HTML, CSS
Software: Microsoft PowerPoint, Microsoft Excel, Altium, TINA-TI, OpenRocket, Blender, Adobe Premiere Pro